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Факультет «Информатика и системы управления»
Кафедра ИУ5 «Системы обработки информации и управления»

Курс «Парадигмы и конструкции языков программирования»
Отчет по рубежному контролю №1
«Вариант А. 15»

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Листинг кода:

```
1  from dataclasses import dataclass, field
2  from typing import Optional
3  import random
4
5
6  @dataclass 4 usages
7  class File:
8      file_id: int
9      file_size: int
10     folder_id: Optional[int] = None # связь 1→M
11
12
13  @dataclass 5 usages
14  class Folder:
15      folder_id: int
16      name: str
17      files: list[File] = field(default_factory=list) # 1→M
18
19
20  @dataclass 3 usages
21  class FolderFileLink:
22      folder_id: int
23      file_id: int
24
25
26  def generate_data(num_folders: int = 5, files_per_folder: int = 5): 1 usage
27      folders: list[Folder] = []
28      files: dict[int, File] = {}
29      links: list[FolderFileLink] = []
30
```

```
31  file_counter = 1
32
33  for i in range(1, num_folders + 1):
34      folder = Folder(folder_id=i, name=f"Folder_{i}")
35      for j in range(1, files_per_folder + 1):
36          f = File(file_id=file_counter, file_size=random.randint(a=100, b=10_000), folder_id=i)
37          folder.files.append(f)
38          files[file_counter] = f
39          file_counter += 1
40      folders.append(folder)
41
42  for _ in range(10):
43      folder = random.choice(folders)
44      file = random.choice(list(files.values()))
45      link = FolderFileLink(folder_id=folder.folder_id, file_id=file.file_id)
46      if link not in links:
47          links.append(link)
48
49  return folders, files, links
50
51
52  def query_all_folders_and_files(folders: list[Folder]): 1 usage
53      print("=== 3anpoc A1 ===\n")
54      for folder in sorted(folders, key=lambda f: f.name):
55          print(f"{folder.name}")
56          for f in folder.files:
57              print(f"File {f.file_id} (size={f.file_size})")
58      print()
```

```

61 def query_folders_by_total_size(folders: list[Folder]): 1 usage
62     print("=== 3anpoc A2 ===\n")
63     totals = [(folder, sum(f.file_size for f in folder.files)) for folder in folders]
64     totals.sort(key=lambda x: x[1], reverse=True)
65     for folder, total in totals:
66         print(f"{folder.name}: total size = {total}")
67     print()
68
69
70 def query_even_folders_mn(folders: list[Folder], files: dict[int, File], links: list[FolderFileLink]): 1 usage
71     print("=== 3anpoc A3 ===\n")
72     even_folders = [
73         f for f in folders
74         if any(ch.isdigit() and int(ch) % 2 == 0 for ch in f.name)
75     ]
76
77     for folder in even_folders:
78         linked_file_ids = [link.file_id for link in links if link.folder_id == folder.folder_id]
79         linked_files = [files[fid] for fid in linked_file_ids]
80
81         print(f"{folder.name}")
82         if linked_files:
83             for f in linked_files:
84                 print(f"Linked File {f.file_id} (size={f.file_size})")
85         else:
86             print("Нет связанных файлов")
87         print()

```

```

89
90  if __name__ == "__main__":
91     folders, files, links = generate_data()
92
93     query_all_folders_and_files(folders)
94     query_folders_by_total_size(folders)
95     query_even_folders_mn(folders, files, links)
96

```

Результат исполнения:

```
C:\Users\Mark\AppData\Local\Programs\Python\Python313\python.exe C:\Users\Mark\PycharmProjects\ROSTININ-IUS-PCPL\main.py
=== 3anpoc A1 ===

Folder_1
File 1 (size=1119)
File 2 (size=9608)
File 3 (size=2477)
File 4 (size=9593)
File 5 (size=8153)

Folder_2
File 6 (size=5181)
File 7 (size=3748)
File 8 (size=1728)
File 9 (size=4106)
File 10 (size=7431)

Folder_3
File 11 (size=3315)
File 12 (size=3822)
File 13 (size=7296)
File 14 (size=6672)
File 15 (size=8678)

Folder_4
File 16 (size=3970)
File 17 (size=1720)
File 18 (size=2289)
File 19 (size=9051)
```

File 20 (size=7291)

Folder_5

File 21 (size=1605)

File 22 (size=7807)

File 23 (size=6025)

File 24 (size=9457)

File 25 (size=668)

=== 3anpoc A2 ===

Folder_1: total size = 30950

Folder_3: total size = 29783

Folder_5: total size = 25562

Folder_4: total size = 24321

Folder_2: total size = 22194

=== 3anpoc A3 ===

Folder_2

Linked File 10 (size=7431)

Linked File 24 (size=9457)

Folder_4

Linked File 16 (size=3970)

Linked File 13 (size=7296)

Process finished with exit code 0